

Homework 6

1 Question 6

(a) $\forall x P(x) \vdash \forall y P(y)$

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|----|------------------|------------------|
| 1. | $\forall x P(x)$ | premise |
| 2. | y_0 | fresh variable |
| 3. | $P(y_0)$ | $\forall e, 1$ |
| 4. | $\forall y P(y)$ | $\forall i, 2-3$ |

(b) $\forall x (P(x) \rightarrow Q(x)) \vdash (\forall x \neg Q(x)) \rightarrow (\forall x \neg P(x))$

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|-----|---|-----------------------|
| 1. | $\forall x (P(x) \rightarrow Q(x))$ | premise |
| 2. | $\forall x \neg Q(x)$ | assumption |
| 3. | x_0 | fresh variable |
| 4. | $P(x_0) \rightarrow Q(x_0)$ | $\forall e, 1$ |
| 5. | $\neg Q(x_0)$ | $\forall e, 2$ |
| 6. | $P(x_0)$ | assumption |
| 7. | $Q(x_0)$ | $\rightarrow e, 4, 6$ |
| 8. | $\perp$ | $\neg e, 5, 7$ |
| 9. | $\neg P(x_0)$ | $\neg i, 6-8$ |
| 10. | $\forall x \neg P(x)$ | $\forall i, 3-9$ |
| 11. | $(\forall x \neg Q(x)) \rightarrow (\forall x \neg P(x))$ | $\rightarrow i, 2-10$ |

(c) $\forall x (P(x) \rightarrow \neg Q(x)) \vdash \neg(\exists x (P(x) \wedge Q(x)))$

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|-----|--|-----------------------|
| 1. | $\forall x (P(x) \rightarrow \neg Q(x))$ | premise |
| 2. | $\exists x (P(x) \wedge Q(x))$ | assumption |
| 3. | x_0 | |
| 4. | $P(x_0) \wedge Q(x_0)$ | $\exists e, 2, 3$ |
| 5. | $P(x_0) \rightarrow \neg Q(x_0)$ | $\forall e, 1$ |
| 6. | $P(x_0)$ | $\wedge e_1, 4$ |
| 7. | $Q(x_0)$ | $\wedge e_2, 4$ |
| 8. | $\neg Q(x_0)$ | $\rightarrow e, 5, 6$ |
| 9. | $\perp$ | $\neg e, 7, 8$ |
| 10. | $\neg(\exists x (P(x) \wedge Q(x)))$ | $\neg i, 2-9$ |

2 Question 9

(b) $S \rightarrow \exists x Q(x) \vdash \exists x (S \rightarrow Q(x))$

1.	$S \rightarrow \exists x Q(x)$	premise
2.	$S \vee \neg S$	LEM
3.	S	assumption
4.	$\exists x Q(x)$	$\rightarrow$ e, 1, 3
5.	x_0	fresh variable
6.	$Q(x_0)$	assumption
7.	S	assumption
8.	$Q(x_0)$	reit., 6
9.	$S \rightarrow Q(x_0)$	$\rightarrow$ i, 6–7
10.	$\exists x (S \rightarrow Q(x))$	$\exists$ i, 8
11.	$\exists x (S \rightarrow Q(x))$	$\exists$ e, 4, 5–9
12.	$\neg S$	assumption
13.	x_0	fresh variable
14.	S	assumption
15.	$\perp$	$\neg$ e, 11, 13
16.	$Q(x_0)$	$\perp$ e, 14
17.	$S \rightarrow Q(x_0)$	$\rightarrow$ i, 13–15
18.	$\exists x (S \rightarrow Q(x))$	$\exists$ i, 16
19.	$\exists x (S \rightarrow Q(x))$	reit., 17
20.	$\exists x (S \rightarrow Q(x))$	$\vee$ e, 2, 3–10, 11–18
21.	$\exists x (S \rightarrow Q(x))$	reit., 19